

RPD SANGATHAN

Flutter Field App Feature Guide

Screen-wise documentation of features, fields,
validation, APIs, access rules, and offline behaviour.

Stack: Flutter | GetX | Hive | Dio

Platforms: Android and iOS

Languages: English, Hindi, Bhojpuri

1. How the app works

RPD is a field membership app for karyakartas. A user signs in with a mobile OTP, completes a 3-step join form, then uses Home, Activity, Members, and More. The app talks to the RPD backend over HTTPS and keeps a local Hive cache so Home, posts, and drafts still appear when the phone is offline.

Startup

- Boot reads assets/env for API_BASE_URL. If missing, the API Settings screen opens.
- If no access token exists, the user goes to Mobile login.
- If a token exists, the app calls GET /auth/me. Success opens Home or Join. Failure signs the user out.
- Expired tokens try POST /auth/refresh. If refresh fails, the session is cleared and login opens quietly.

Sign-in

- User enters a 10-digit Indian mobile (starts with 6-9). App sends POST /auth/otp/request on WhatsApp channel.
- In development, OTP is 123456. The app does not prefill it.
- POST /auth/otp/verify returns tokens (90 days) and the member profile. A DRAFT member is created if the number is new.

Join (3 steps, then contribution)

- Step 1 About you: full name, date of birth (age 18-120), gender, optional photo.
- Step 2 Your details: state, district, assembly, optional pincode and address, pledge toggle.
- Step 3 Consent: required membership consent plus optional WhatsApp updates. Submit calls POST /members/register.
- A contribution dialog then asks Primary Member or Volunteer and saves POST /members/contribution.
- Save & exit on steps 1-2 stores the draft in Hive and opens Home without finishing join.

Verified vs unverified

- Verified users (status VERIFIED, or Super Admin) can use Activity, create post, add member, and the membership card.
- Unverified users can stay on Home, open More, view posts, and edit Profile. Gated actions send them back to Join/status.

Super Admin

- Member flag isSuperAdmin (or post SUPER_ADMIN) is treated as verified.
- They can open every gated screen, see all region posts, and use Verification inbox for all areas.

Member ID

- Each member has a table row id (1, 2, 3...). The card and profile show it as RPD-1, RPD-2, and so on.

Local storage (Hive)

settings: locale, API URL, tokens. user: profile. draft: join form. booths: cached booths. region_posts: post list. sync_queue: pending photo posts and activities. Sign out clears tokens, profile, draft, posts, queue, recruits, activities, and tasks. Locale and API URL stay.

Languages

EN, HI, and BHO. The dropdown is on Login, Home, Activity, Members, and More. Choice is saved in Hive.

2. App structure

Layer	What it does
GetX routes	Named screens in app_pages.dart. Shell holds Home, Activity, Members, More.
SessionController	Auth, profile, home load, location, join/register, gates, post sync.
ApiClient	Dio client. Adds Bearer token. Refreshes on 401. Signs out if refresh fails.

Layer	What it does
HiveService	Offline cache and drafts.
Backend	Express + Prisma + Postgres. Media in MinIO/S3.

Bottom navigation

Tab	Opens	Gated?
Home	Dashboard	No
Activity	Record meeting / griha / programme / training	Yes - verified only
Center FAB	Create post	Yes - verified only
Members	My recruits	View open; Add member gated
More	Card, posts, tasks, meeting, language, sign out	Card overlay gated

3. Screens - fields and working

3.1 Boot

Route / (Routes.boot)
File lib/features/boot/boot_view.dart
Purpose Startup router. No user fields.

Fields and controls

Field / control	Type	Working
Spinner	Loading	Waits while session is restored

How it works

- If API URL is missing, go to API Settings.
- If no token, go to Mobile login.
- If token is valid, go to Home (verified) or Join/status (unverified).
- If token is invalid, clear session and go to login.

Access Everyone at launch
APIs GET /auth/me, POST /home/active
Offline Uses cached profile if the network call fails for a non-401 reason.

3.2 Mobile login

Route /mobile
File lib/features/auth/auth_views.dart
Purpose Collect the mobile number and send OTP.

Fields and controls

Field / control	Type	Working
Language	Dropdown EN/Hi/BHO	Changes app language immediately
Your mobile number	Heading	Screen title
Help text	Text	Explains that a 6-digit code will be sent
Mobile number	Text +91, 10 digits	Must match 6-9 then 9 digits
Send code	Button	POST /auth/otp/request then opens OTP screen
Terms line	Footer	Membership terms and privacy notice

How it works

- Invalid number shows: Enter a 10-digit Indian mobile number.
- A new number creates a DRAFT member after OTP verify.
- Changing the number later clears a previous join draft.

Access Signed-out users
APIs POST /auth/otp/request { mobile, channel: WHATSAPP }
Offline Needs network.

3.3 OTP

Route /otp
File lib/features/auth/auth_views.dart

Purpose Enter the 6-digit code.

Fields and controls

Field / control	Type	Working
Back	Icon	Return to mobile screen
Enter the code	Heading	
+91 number change	Link	Goes back to change the number
OTP boxes	6 digits	Continue enables at 6 characters
Continue	Button	POST /auth/otp/verify; saves tokens and profile
Resend / SMS	Text	Shown only; not wired to a new request

How it works

- Demo OTP is 123456 on the development API.
- On success: verified users open Home; new/draft users open Join or member status.
- Super Admin skips join and opens Home.

Access After Send code

APIs POST /auth/otp/verify { mobile, code }

Offline Needs network.

3.4 Join step 1 - About you

Route /join/personal

File lib/features/join/join_views.dart

Purpose Personal details. Also used inside Member status when the application is still a draft.

Fields and controls

Field / control	Type	Working
Save & exit	App bar	Saves draft to Hive and opens Home
Step 1 of 3	Progress	Visual only
Photo	Circle + camera	Optional. Camera or Gallery; upload POST /members/photo
Full name	Text	Required, at least 2 characters
Date of birth *	Date picker	Age must be 18 to 120 DD-MM-YYYY
Gender	Dropdown	Male / Female / Other
Continue	Button	Validates, marks step 1 done, opens step 2

How it works

- Values are written to the Hive draft as the user types.
- Photo is optional. Tap the photo later to view it full screen.

Access New or unverified members

APIs POST /members/photo (optional)

Offline Draft stays on the device. Photo upload needs network.

3.5 Join step 2 - Your details

Route /join/booth

File lib/features/join/join_views.dart

Purpose Address and constituency.

Fields and controls

Field / control	Type	Working
Back	App bar	Returns to step 1
Save & exit	App bar	Opens Home with draft saved
State *	Dropdown	GET /geo/states (union territories excluded)
District *	Dropdown	Loads after state: GET /geo/districts
Assembly constituency *	Dropdown	Loads after district; first booth is auto-assigned
Pincode	6 digits	Optional; if filled must be 6 digits
Address	Text	Optional
Pledge / updates	Checkbox	Default on. Stored as pledgeUpdates
Continue	Button	Opens Consent

How it works

- Back arrow uses Get.offNamed to personal so the stack stays clean.
- Assembly pick loads booths for that AC and stores the first booth id in the draft.

Access Join flow

APIs GET /geo/states, /geo/districts, /geo/assemblies, GET /booths?assemblyId=

Offline Draft is local. Empty dropdowns if geo APIs fail.

3.6 Join step 3 - Consent

Route /join/consent

File lib/features/join/join_views.dart

Purpose Legal consent and registration.

Fields and controls

Field / control	Type	Working
What we collect, and why	Read-only	Name, mobile, DOB, address, booth, photograph
I agree to the above	Required checkbox	Must be checked
WhatsApp updates	Optional checkbox	whatsappOptIn
Submit application	Button	POST /members/register then contribution dialog

How it works

- Sends fullName, dateOfBirth, gender, boothId or assemblyId, address, pincode, locale, consent version, WhatsApp flag.
- On success the member becomes VERIFIED and gets membership number RPD-{rowId}.
- Contribution dialog is shown next (Primary Member or Volunteer).

Access Join flow

APIs POST /members/register

Offline Submit needs network.

3.7 Contribution dialog

Route Modal after register

File lib/features/join/contribute_dialog.dart

Purpose How the member wants to contribute.

Fields and controls

Field / control	Type	Working
Close	X	Dismiss without saving
Primary Member	Radio	Shows optional referral code / phone
Volunteer	Radio	Shows Online/Offline and weekly hours 3-40
Next	Button	POST /members/contribution then Home

How it works

- Primary Member: type PRIMARY_MEMBER, optional referralCode.
- Volunteer: type VOLUNTEER, volunteerMode ONLINE or OFFLINE, weeklyHours 3 to 40.

Access After successful register

APIs POST /members/contribution

Offline Save needs network. Close always works.

3.8 Member status

Route /verification/status

File lib/features/verification/member_status_view.dart

Purpose Resume join if draft/rejected; otherwise show application status.

Fields and controls

Field / control	Type	Working
About you form	Embedded	Shown when DRAFT, REJECTED, or name is empty
Status card	Read-only	Pending / Suspended / Withdrawn / Verified copy
Check status	Button	GET /auth/me; opens Home if verified
Sign out	Button	Logout dialog

How it works

- Unverified users land here when they tap a gated action.
- Pending copy explains the application is with the Mandal President.

Access Unverified members

APIs GET /auth/me

Offline Shows cached status; Check status needs network.

3.9 Home

Route	/shell tab Home
File	lib/features/home/home_view.dart
Purpose	Dashboard after login.

Fields and controls

Field / control	Type	Working
Avatar	Tap	Opens Profile
Greeting + name	Text	Good morning/afternoon/evening + full name. Mobile is not shown.
Language pill	Dropdown	EN / HI / BHO
Notifications bell	Tap	Opens Notifications (static list)
Your community score	Number /100	From home stats or booth healthScore
View booth	Link	Opens Booth health
Queue banner	If syncCount > 0	Opens Sync queue
Not verified banner	If unverified	Opens Join/status
Tasks due	Banner	Opens Tasks
Upcoming events	Card + Join	From GET /home
Members added / Meetings held	Stats	From GET /home
Mandal rank / points	Card	Leaderboard snapshot
Recent videos	Grid	See more opens full list; tap opens YouTube URL
Recent blogs	Rail	See more opens full list
Recent activity near you	Rail	From home nearby cards
Last activity	Card	Opens Activity tab if verified
Pull to refresh	Gesture	Reloads GET /home and pending posts

How it works

- Home loads GET /home on first open and after refresh.
- Unverified users can stay here after Save & exit.

Access	All signed-in users
APIs	GET /home
Offline	Shows last cached home payload.

3.10 Home lists and video

Route	/home/videos, /home/blogs, /home/nearby, /home/events, /video
File	lib/features/home/home_feed_list_view.dart, youtube_player_view.dart
Purpose	Full lists of Home sections.

Fields and controls

Field / control	Type	Working
App bar title	Text	Section name
Feed cards	List	Image, title, place/source
Join	Button on events	UI only from cached home data

How it works

- Lists reuse SessionController.home. No extra API.
- /video embeds youtube.com/embed/{id}. Home currently opens the external URL instead.

Access	All signed-in users
APIs	None (cached /home)
Offline	Uses last Home load.

3.11 Activity hub

Route	/shell tab Activity
File	lib/features/work/work_view.dart
Purpose	Choose what to record.

Fields and controls

Field / control	Type	Working
Meeting	Tile	Opens activity capture with type MEETING
Griha sampark	Tile	Type GRIHA_SAMPARK
Programme	Tile	Type PUBLIC_PROGRAMME
Training	Tile	Type TRAINING

How it works

- Unverified users are sent to Join/status immediately.
- Add member and Something else were removed from this hub.

Access	Verified only
APIs	None on this screen
Offline	N/A

3.12 Activity capture

Route	/activity/details
File	lib/features/activity/activity_views.dart
Purpose	Record an activity with photos and place.

Fields and controls

Field / control	Type	Working
Purpose	Read-only	Selected activity type
Add photos	Camera / Gallery	At least one photo required
Time	Read-only	Captured now
Location	Lat/long or retry	Uses device GPS; tap to retry
Place name	Text	Village, school or landmark; at least 2 characters
Submit	Button	POST /activities if online; else Hive + sync queue

How it works

- On success the Activity saved screen shows queue and pending points.
- Confirm and Rejected screens exist as older/demo routes.

Access	Verified only
APIs	POST /activities
Offline	Saved locally with status SAVED_LOCAL and queued.

3.13 Members - My recruits

Route	/shell tab Members
File	lib/features/members/members_view.dart
Purpose	People this user added.

Fields and controls

Field / control	Type	Working
Verified / Pending / Rejected	Count chips	From GET /members/recruits
Add member	Button	Verified only; opens mobile check
Recruit row	Name, booth, status	List item

How it works

- Add member step 1: 10-digit mobile, GET /members/check/{mobile}.
- If already a member, shows membership number and name.
- Step 2 Recruit consent: full name, DOB, gender, required agree, 6-digit OTP field (length checked locally), POST /members/recruit.

Access	View: all signed-in. Add: verified.
APIs	GET /members/recruits, GET /members/check/:mobile, POST /members/recruit
Offline	Shows last loaded list.

3.14 More

Route	/shell tab More
File	lib/features/more/more_view.dart
Purpose	Secondary destinations.

Fields and controls

Field / control	Type	Working
Verification inbox	Card	Only Super Admin or office-bearer posts
Posts in your region	Card	Opens post list
Membership card	Card	Verified overlay; flip and share
Booth health	Card	Score components for assigned booth
My tasks	Card	GET /tasks grouped overdue / today / week
Booth meeting	Card	GET /meetings; check-in is a placeholder
Language	Dropdown	EN / HI / BHO
Sign out	Button	Confirm dialog, then login

How it works

- Membership card is an overlay, not a full-page route.
- Card front: name, Member ID RPD-n, mobile, referral, post, area, valid thru.
- Card back: address, pincode, state, district, AC name, issued date. No AC on the front.

Access	Signed-in; card and some actions gated
APIs	Depends on destination
Offline	Card uses cached profile if refresh fails.

3.15 Profile

Route /profile
File lib/features/profile/profile_view.dart
Purpose View and edit the signed-in member.

Fields and controls

Field / control	Type	Working
Photo	Tap	Camera or Gallery; POST /members/photo
Member ID	Read-only	RPD-{rowId}
Mobile number	Read-only	Last 10 digits
Full name	Text	Required, 2+ characters
Date of birth *	Date picker	Age 18-120
Gender	Dropdown	Male / Female / Other
Address	Text	Optional
Pincode	6 digits	Optional; 6 digits if filled
State / District / Assembly *	Cascading	Same geo APIs as join
Save changes	Button	PATCH /members/me

How it works

- Opened from the Home avatar. Not verification-gated.
- RefreshMe runs on open so Super Admin and new IDs appear after re-login.

Access All signed-in users
APIs GET /auth/me, geo APIs, PATCH /members/me, POST /members/photo
Offline Display works from cache; save needs network.

3.16 Region posts

Route /posts and /posts/create and /posts/video
File lib/features/post/post_views.dart
Purpose Share photo, audio, or video with the region.

Fields and controls

Field / control	Type	Working
My posts / Other posts	Tabs	Split by authorId and mobile
Post card media	Image / audio / video thumb	Video shows thumbnail + play icon; tap opens player
Description	Text under media	Optional
Author time	Footer	Name and relative time only
Create FAB	Orange post icon	Verified only
Photo / Audio / Video	Tabs on create	Photo works offline; audio/video need network
Camera / Gallery or Record / Upload	Source cards	Horizontal; Replace keeps current media until a new file is picked
Description (optional)	Growing text field	Max 2000 on API
Post update	Button	Uploads multipart file + mediaType IMAGE VIDEO AUDIO

How it works

- Video: a JPEG thumbnail is generated on the phone and sent as field thumbnail. Stored as thumbnail_key.

- Photo offline: saved in Hive as pending REGION_POST and uploaded later from Sync or Home load.
- List does not start VideoPlayer on every card.

Access View: all. Create: verified.

APIs GET /posts, POST /posts (multipart)

Offline List from Hive. Photo compose offline. Audio/video blocked offline.

3.17 Verification inbox

Route /verification

File lib/features/verification/inbox_view.dart

Purpose Office bearers accept pending activities.

Fields and controls

Field / control	Type	Working
To check	App bar	
Activity card	Type, booth, actor	From GET /verification/inbox
Accept	Button	POST /verification/{id}/accept
Ask again	Button	Not wired
Locked message	Text	Shown if the user has no verification post

How it works

- Allowed posts: Booth Adhyaksh and above, plus Super Admin.
- Super Admin sees all pending items, not only one mandal.

Access Verification posts or Super Admin

APIs GET /verification/inbox, POST /verification/:id/accept

Offline Needs network to load.

3.18 Other screens

Route See table

File tasks, booth, meeting, sync, notifications, settings

Purpose Supporting tools.

Fields and controls

Field / control	Type	Working
My tasks /tasks	List	GET /tasks: overdue, today, this week
Booth health /booth/health	Scores	GET /booths/{id}/health; Appoint Panna is not wired
Booth meeting /meeting	Agenda + invitees	GET /meetings; Start check-in opens placeholder QR
Check-in /meeting/checkin	Placeholder	No live QR scan
Sync /sync	Queue	Try uploading now retries pending photo posts
Notifications /notifications	Static list	Mark all read is local only
API Settings /settings/api	URL field	Required if assets/env has no API_BASE_URL
Error log /settings/logs	Log list	In-memory AppLog; no menu link

How it works

- Notifications and meeting check-in are UI placeholders.
- Sync queue also appears as a Home banner when items are waiting.

Access Signed-in unless noted

APIs	See field column
Offline	Queue and logs are local.

4. Access matrix

Feature	Unverified	Verified member	Super Admin / office bearer
Home	Yes	Yes	Yes
Save & exit join	Yes	n/a	n/a
Activity tab / capture	No - sent to join	Yes	Yes
Create post FAB	No	Yes	Yes
View region posts	Yes	Yes	Yes (all posts)
Add member	No	Yes	Yes
Membership card	No	Yes	Yes
Profile	Yes	Yes	Yes
Verification inbox	No	Only listed posts	Yes, all areas
Tasks / meeting / booth	Yes	Yes	Yes

Office-bearer posts that can open Verification inbox: SUPER_ADMIN, BOOTH_ADHYAKSH, MANDAL_PRESIDENT, ASSEMBLY_IN_CHARGE, DISTRICT_SECRETARY, DISTRICT_GENERAL_SECRETARY, DISTRICT_PRESIDENT, REGIONAL_PRESIDENT, STATE_GENERAL_SECRETARY, STATE_PRESIDENT, NATIONAL_GENERAL_SECRETARY, NATIONAL_PRESIDENT.

5. API map

Method	Path	Used from
POST	/auth/otp/request	Mobile login
POST	/auth/otp/verify	OTP
POST	/auth/refresh	ApiClient on 401
GET	/auth/me	Boot, Profile, Card
POST	/home/active	Boot
GET	/home	Home
GET	/geo/states districts assemblies	Join, Profile
GET	/booths	Join auto-booth
GET	/booths/nearby	Nearby booth helper
GET	/booths/{id}/health	Booth health
POST	/members/register	Consent submit
POST	/members/contribution	Contribution dialog
POST	/members/photo	Join/Profile photo
PATCH	/members/me	Profile save
GET	/members/check/{mobile}	Add member
POST	/members/recruit	Recruit consent
GET	/members/recruits	Members tab
POST	/activities	Activity capture
GET/POST	/posts	Region posts; video includes thumbnail
GET	/tasks	Tasks
GET	/meetings	Meeting
GET	/verification/inbox	Inbox

Method	Path	Used from
POST	/verification/{id}/accept	Accept activity

6. Typical journeys

New karyakarta

- Login with mobile + OTP 123456 (dev) -> About you -> Details -> Consent -> Contribution -> Home.
- Member ID on the card becomes RPD-n from the members table row.

Unverified browsing

- Save & exit after step 1 or 2 -> Home with Not verified banner.
- Tapping Activity, FAB, card, or Add member returns them to Join/status.

Create a video post

- Verified user -> FAB -> Video tab -> Record or Upload (needs signal).
- App builds a JPEG thumbnail, sends file + thumbnail + mediaType VIDEO.
- List shows thumbnail with a play icon. Tap opens the in-app player.

Create a photo post offline

- Photo tab works without network. Post is stored in Hive and the sync queue.
- When the phone is online, Home load or Sync -> Try uploading now sends it.

Super Admin

- Sign out and sign in again after the flag is set so /auth/me returns isSuperAdmin.
- Home and all tabs unlock. More shows Verification inbox. Posts list is not limited by district.

7. Notes

- Notifications and meeting QR check-in are placeholder UI.
- Activity Confirm/Rejected routes are older demo screens.
- Error log and YouTube player routes exist but are not linked from More/Home.
- After a database wipe, leftover tokens cause a silent sign-out on the next launch.

This guide matches the Flutter app in rpd_app as of September 2026.